CHEMISTRY ADAPTLY

BONDING – METALLIC BONDING

KEY VOCABULARY

Metallic bond: the strong attraction between positive metal ions and delocalised electrons.

Delocalised: electrons that are free to move through the whole structure.

Lattice: a regular, repeating arrangement of ions.

Malleable: can be hammered or bent into shape without breaking.

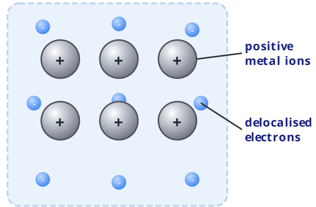
COMMON MISCONCEPTIONS

~~Metals are made of molecules.~~
✓ Metals are a giant lattice of ions.

~~The electrons belong to one atom.~~
✓ Electrons are delocalised (shared by all).

~~Metals conduct because ions move.~~
✓ The delocalised electrons carry the charge.

THE METALLIC LATTICE



Positive **metal ions** sit in a 'sea' of **delocalised electrons**.

1 DEFINE THE BOND

Describe what is meant by **metallic bonding**. **[2 marks]**

Use the words 'positive metal ions' and 'delocalised electrons'.

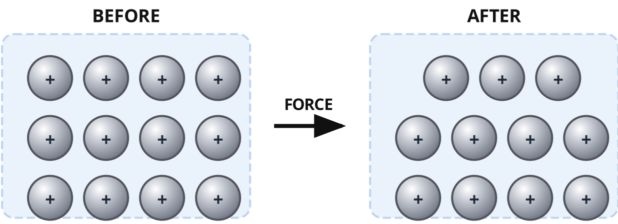
2 ELECTRICAL CONDUCTIVITY

Explain why metals are good conductors of electricity. **[2 marks]**

Think about which particles are free to move.

3 MALLEABILITY

The diagram shows the lattice before and after a force is applied. **Explain** why the layers of ions can slide. **[2 marks]**



4 MELTING POINT

Complete the sentence using the word bank. **[3 marks]**

Metals have _____ melting points because there is a _____ electrostatic _____ between the ions and the delocalised electrons.

high | strong | attraction

5 LABEL THE PARTICLES

In the lattice diagram, what do the **large '+' circles** and the **small '-' circles** represent? **[2 marks]**

6 WHY SO STRONG?

Suggest why a metal with ions of charge 2+ has a higher melting point than one with 1+ ions. **[2 marks]**

TIP: *Positive metal ions are held in a 'sea' of delocalised electrons.*

✓ Think: Why does a bigger ionic charge give a stronger metallic bond?